

Deep Transfer Learning for Schizophrenia Detection using Brain MRI

A Minor Project Report

Submitted in partial fulfillment of the
Requirements of VII-Semester for the degree

of

Bachelor of Technology

in

COMPUTER SCIENCE & ENGINEERING

by

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NOVEMBER 2022

DECLARATION

We hereby declare that the work described in this thesis, entitled “**Deep Transfer Learning for Schizophrenia Detection using Brain MRI**” which is being submitted by us in partial fulfillment for the VII Semester of the degree of Bachelor of Technology in the Department of **Computer Science and Engineering** to the National Institute of Technology Raipur is the result of investigations carried out by us under the guidance of **Dr. Dilip Singh Sisodia, Associate Professor, Department of Computer Science & Engineering, NIT Raipur**.

The work is original and has not been submitted for any Degree/Diploma of this or any other Institute/university.

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CERTIFICATE

This is to certify that the project entitled “**Deep Transfer Learning for Schizophrenia Detection using Brain MRI**”, which is being submitted by Amitesh Agrawal (Roll no. 19115086), Siddhant Mudholkar (Roll no. 19115086) in partial fulfillment for VII-semester of the degree of **Bachelor of Technology in Computer Science & Engineering** to National Institute of Technology Raipur is a record of bonafide work carried out by them under my guidance and supervision.

The matter presented in this project document has not been submitted by them for the award of any other degree elsewhere.

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ACKNOWLEDGEMENT

We would like to acknowledge our college **National Institute of Technology, Raipur** for providing a holistic environment that nurtures creativity and research-based activities.

We express our sincere thanks to **Dr. Dilip Singh Sisodia, Associate Professor, Department of Computer Science & Engineering, NIT Raipur**, the guide of the project for guiding and **correcting** throughout the process with attention and care. He has frequently suggested us creative ideas and guided us through major hurdles that occurred during the duration of the project.

We would also thank **Dr. Dilip Singh Sisodia, Head of the Department**, and all the faculty members without whom this project would be a distant reality. We also extend our heartfelt thanks to our family and friends who supported us.

Thank You!!

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ABSTRACT

Schizophrenia is a severe mental disorder in which people experience a combination of hallucinations, delusions, and extremely disorganized thinking. Early diagnosis and treatment can reduce family difficulties and lower social costs. Its detection takes a long time as it requires several interviews with psychiatrists and taking MRI images of the brain. Medically no objective assessment index has yet been developed for schizophrenia. The development of machine learning and deep learning algorithms helps in diagnosing diseases easily. In machine learning models, handcrafted features are used, which require domain experts to build the feature set, which takes a lot of time and expertise. While Deep Learning (DL) models use brain MRI images to detect schizophrenia, a huge dataset is needed to train the model, increasing the computation time. Also, only a few MRI dataset is publicly available for schizophrenia detection. Recently transfer learning-based DL models have been trained on three views (sagittal, coronal, and axial) of 3D MRI images. However, the features extracted by these models are noisy, affecting the model's overall performance. Hence, in this work, we are using the axial view of brain images which focuses more on contributing features to train transfer learning-based VGG19 model for the detection of schizophrenia. A publicly available dataset of brain MRI images published by COBRE is used in this work. The dataset consists of 146 patients' brain MRIs. These MRI images are in 3D form; we did data analysis on the 3D image to get slice of axial view of the brain. Axial view image is preprocessed through thresholding for better differentiability between intensities. Since data is less, we are using data augmentation techniques which also reduces overfitting. We used a VGG-19 pre-trained network and added some fully connected layers for proposed model. ReLU used for hidden layers and sigmoid used for last layer as activation function, Adam optimizer for model optimization. We evaluated our model using standard metrics such as accuracy, precision, recall, and F1 score. The proposed model obtained 90.9% accuracy over the dataset. It is further compared with other standard classification models using standard metrics. The result shows that we have improved accuracy by maximum 3% and minimum 1% over the existing models.

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CHAPTER 1

INTRODUCTION

1.1. Overview

Schizophrenia is a very serious medical illness which affects the behavior and thinking of person suffering from it. Symptoms of schizophrenia can be of different types like psychotic symptoms which changes the thinking of the person, it includes hallucinations and delusions, negative symptoms which includes loss of motivation or enjoyment in daily chores, difficulty in expressing emotions and moving away from social life, cognitive symptoms which creates different problems like lack of concentration and weak memory. It is found in 1 out of 300 people in the world [1].

With every year there is an increase in the number of patients facing problems due to schizophrenia as we can see from the graph in Figure 1 shows the growth and number of patients suffering from schizophrenia in thousands from year 1996 to 2020 in Japan. [2]



Figure 1: Number of patients receiving treatments for schizophrenia, from 1996 to 2020, by gender in Japan [2]

Schizophrenia is one of the top 10 global causes of disability in worldwide nations. The main issue is the mortality rate of this disease, mortality rate in schizophrenia is about 2.5 which leads to a reduction in life expectancy between 15 and 20 years [3]. As lot of people commit suicide due to unbearable hallucinations and delusions. It is a very severe, protracted mental illness. Numerous diverse psychological symptoms are brought on by it. It gets difficult for the patients to differentiate between the real world and hallucinations they are having.

It is one of the costliest diseases to treat and detection of schizophrenia in an early stage can save lot of time and money. Schizophrenia often develops between the ages of 15 and 25 in early adulthood this can be observed in Figure 2. While most males develop symptoms between the ages of 16 and 25, most females experience symptoms several years later, and the incidence of schizophrenia in women is noticeably greater over the age of 30 [4]. For men, the average age of onset is 18, whereas for women it is 25. It is extremely uncommon for those under the age of 10 or those over the age of 40 to develop schizophrenia.

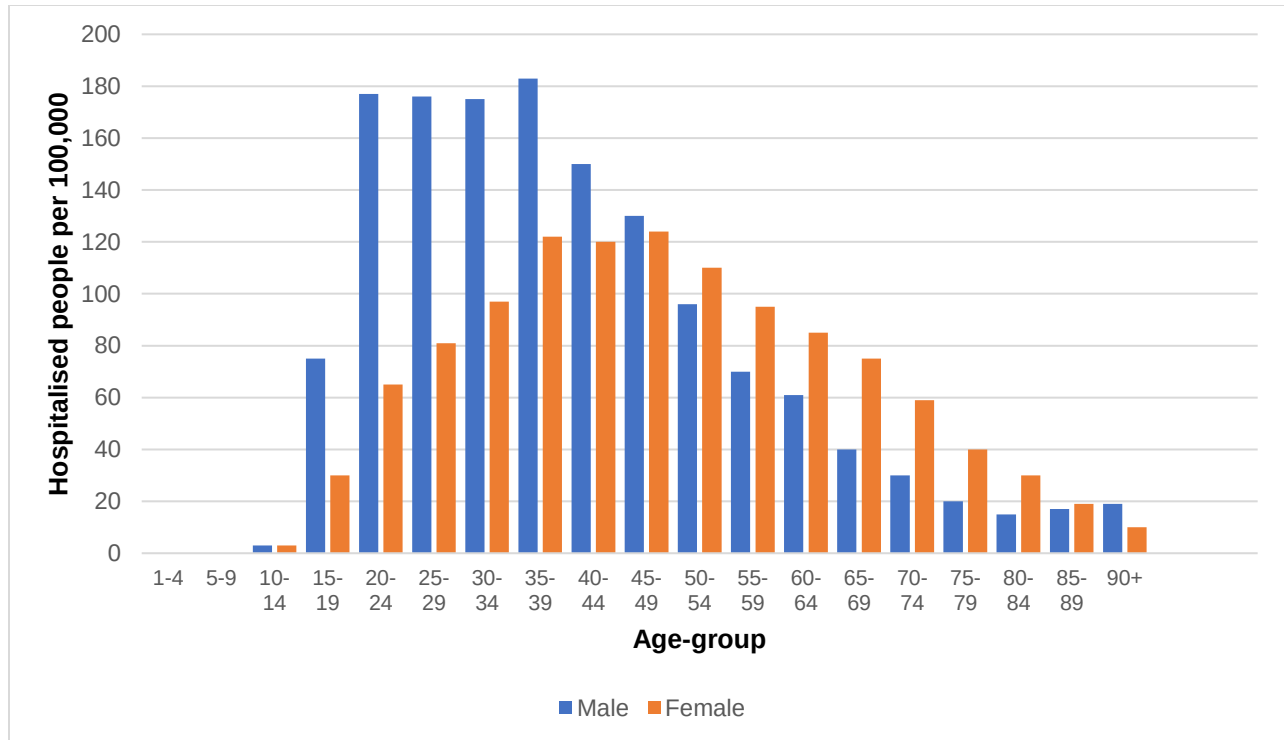


Figure 2 Hospitalizations for schizophrenia per 100,000 by age group [5]

As we can see schizophrenia is increasing every year, and the detection of schizophrenia takes a long time which includes regular clinic visits and interviews with psychiatrists which is based on patients' behavior. Due to the subjective behavior of a doctor's diagnosis, there can be cases that will be missed or misdiagnosed. Thus, there are lots of experiments going on to detect schizophrenia as fast as we can which can help people to save costs on medical bills and time, it can also reduce family burden. As the use of computer-aided diagnosis is used to detect many other diseases currently like brain tumor segmentation[6], Parkinson's disease [7] and Alzheimer's detection [8]. There are many machine learning models which work by taking genetics data as the data with Magnetic Resonance Imaging (MRI) is limited. But now as more data is becoming available, we can directly use MRI data and train a deep learning model which can help to detect schizophrenia faster with more accuracy. Therefore, the detection of this disease can be done early and quickly with brain MRI scans using our deep neural network model, and the diagnosis and treatment can be escalated and the probability of increasing the life span of that patient will increase. Even after using the model, we still need the doctor's assistance as our model is not 100% accurate but still, it will give some hints to the patient about schizophrenia.

1.2. Motivation

Detection of schizophrenia is a very tedious and long task as it takes a lot of MRI scans and doctor visits and interviews with the psychiatrist to finally determine whether the person is suffering from schizophrenia or not. Treatment and diagnosis should be done as soon as possible in young people as they have complete life ahead of them, so treatment should be done early.

The dataset available of brain MRI of schizophrenic patients is very less and it is very hard to train any kind of model from that data, we took this as a challenge and tried different ways of achieving good results with a very minimum dataset. In order to completely solve this problem, we have constructed a deep learning model which can help us in detecting of schizophrenia with just an MRI image. We have trained this model with very less data still able to achieve very good results.

Also, recently one of the NIT Raipur Professor and AIMS Professor together found a way to cure 70% of schizophrenia which inspired us to find a way in detecting schizophrenia with less effort and more accuracy. [9]

1.3. Objective

The main objective

- To only extract the most important features by using the axial view as it shows the subcortical and ventricular areas clearly which contributes the most in detection [10], [11].
- Reduce the number of parameters which will make the computation faster and more accurate.
- To improve the accuracy of detection of schizophrenia which was achieved by using transfer learning with VGG19 and some fully connected layers were added after the transfer learning model so that it can learn from the medical images and predict on the basis of that.
- Comparative study of different latest State of the Art image classification models used for transfer learning, and fine tuning hyperparameters.

1.4. Scope

There are a lot of mental diseases nowadays which have similar characteristics like schizophrenia and need brain MRI images to detect and diagnose. Our model architecture can be used to predict those medical diseases.

CHAPTER 2

LITERATURE OVERVIEW

2.1. Existing work

There is lot of research work already done in this field but it is done using machine learning, deep learning models for this problem statement is very few [12], [13]. Support vector machines was used by more than 55% studies related to neuroimaging studies [14].

Shi et al. [15] proposed a method to categorize schizophrenic (SZs) and healthy controls (HCs), their study implemented a novel method called multimodal imaging and multilevel characterization using multiclassifier (M3) using fMRI and sMRI data, it can also find the regions contributing the most for detection of schizophrenia. This approach had a classification accuracy of 83.49%, with a sensitivity of 68.69%, a specificity of 93.75%, and an AUC of 0.8491. Classification performances increases with different processing methods which shows the importance of preprocessing.

Liu et al. [16] proposed method which requires building unique hierarchical networks, where each node and each edge represent a region of interest (ROI) and, respectively, the correlation between a pair of ROIs. When compared to merely node characteristics, we show that edge features significantly improve SCZ/HC classification performance. In order to further examine classification performance, edge features and node features are combined using a multiple kernel learning architecture. According to the experimental results, their suggested method successfully classifies SCZ/HC with an accuracy of 88.72% and an area under the receiver operating characteristic (ROC) curve (AUC) of 0.9521.

Yassin et al. [17] proposed a machine learning approach, where FreeSurfer image analysis suite version 6.0 was used to process the structural MRI images from all subjects in the same way. Meta-analysis wrapper script to improve Neuroimaging Genetics. This method used 6 different classifiers for classifying SZs and HCs. Support vector machine (SVM) and logistic regression performed the best of the six classifiers to differentiate between the subject groups (LR). The feature groups for cortical thickness and subcortical volume were most helpful for categorization. The clinical indices of ASD were extremely consistent with LR and SVM.

There are many papers available which summarized the research done in field of neuroimaging using deep learning [18]–[20]. In the field of schizophrenia, some of the previous work done in this topic have either used 2D Convolutional Neural Network (CNN) [21] or 3D CNN [22] for learning.

Zheng et al. [21] proposed a model which used fMRI dataset with 2D CNN for detection of schizophrenia which uses an improved VGG-16 network with transfer learning as the dataset was very small. They took the axial view of the brain MRI images and used a pretrained VGG-16 model on ImageNet Dataset and the feature extraction layer of the model was fixed. This model with transfer learning outperformed other models like AlexNet, VGG16, ResNet with accuracy 78.36%, 85.27%, 83.09% respectively. The classification accuracy of fMRI based on VGG-16 is up to 84.3%. While the proposed model in this paper achieved an accuracy of 87.85% in the COBRE dataset, which was a major improvement over the current VGG-16 algorithm.

Latha et al. [23] proposed method to use deep belief networks (DBNs) to analyze the schizophrenic subjects based on ventricle region. They used COBRE dataset to evaluate the performance of their

model. To train the targeted region, the DBN employs a variety of learning methods, including stochastic gradient descent (SGD), adaptive gradient (Adagrad), and root-mean-square propagation (RMSProp). It is then compared to a region of interest (ROI) image that includes the ventricle along with other tissues. When compared to a ROI picture, DBN on segmented ventricle region has the lowest error rate. The segmented ventricle image produced by the suggested method has a high area under the curve (0.899), which amply demonstrates its effectiveness.

Zhang et al. [22] proposed a model which used 3D MRI T1 weighted images directly. They used many different methods of preprocessing, first they used Brain Extractor Tool for getting whole brain image and passed that image on 3D VGG-11 model. The proposed method was to do modification on VGG-11 model and apply squeeze excitation block (SE-VGG-11BN Model). The model proposed in this paper performed better than benchmark models as well as converged faster on validation set. This model when tested on standalone data using whole brain scans got an accuracy of 92.1%. On a different dataset (BrainGluSchi) it got an accuracy of 82.2% which is a major improvement over the benchmark models present. Accuracy achieved on COBRE dataset for T1W (whole head) was 0.867 with sensitivity of 0.897 and specificity 0.861 and AUC 0.938.[24] Accuracy achieved on COBRE dataset for T1W (whole brain) was 0.921 with sensitivity of 0.949 and specificity of 0.946 and AUC 0.987.

Oh J et al. [24] proposed a model which used a different strategy for preprocessing by converting the images into a video where each slice of the transverse section was converted to a frame, and these frames were combined into a video. A three-dimensional convolutional neural network (3DCNN) architecture was used for classifying patients with schizophrenia and normal subjects based on the structural MRI data sets. Overall accuracy was 97% which is one of the highest among 3D CNN models. Results from this model also showed the regional analysis of the brain which is shown in Figure 3. The area which includes the right temporal region (marked as #1) contributed the most to distinguishing between scans from patients with schizophrenia and normal subjects, followed by the right temporoparietal (marked as #2) and left frontal (marked as #4) regions [24].

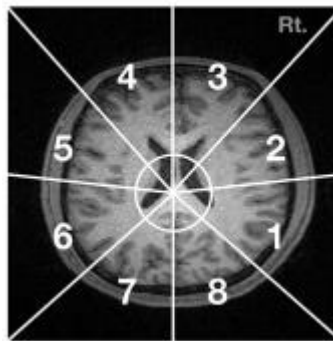


Figure 3 Diagram of eight brain regions [24]

The shortcomings of this proposed model were that 3D CNN has a very high computation cost during training because of high dimensional data which results in a large number of trainable parameters in the model. Secondly the sample dataset used in this paper was relatively small for 3D CNN training. This results in less feature extraction and thus results in lower generalizability of the model. Table 1 summarizes all the literature work which has been done in this field, which

consist dataset used by them, preprocessing technique applied, important remarks from each of the literature work.

Table 1 Summary of literature overview in tabular form

S.No	Author	Dataset Used	Preprocessing	Methods	Remarks
1	Yassin et al. [17]	Private Dataset of 131 patients	FreeSurfer image analysis suite v.6.0	6 different classifier, SVM and LR best results.	Cortical thickness and subcortical volume were most contributing
2	Shi et al. [15]	Center for Biomedical Research Excellence (COBRE)	Data Processing & Analysis of Brain Imaging and data Processing Assistant for Resting-State fMRI	Multimodal imaging and Multilevel characterization using Multiclassifier(M3)	Importance of preprocessing.
3	Liu et al. [16]	Private data of 76 patients	Freesurfer image analysis suite v 5.3.0	Multiple kernel framework	Result varies with region of interest (ROI)
4	Zheng et al. [21]	COBRE	Thresholding, Binarization	2DCNN using Transfer learning	Transfer Learning helps in reduced data
5	Oh J et al. [24]	COBRE, MCICShare, NMorphCH, NUSDAST, BrainGluSchi	Converting the image slices into a video	3DCNN architecture	Different Temporal regions contributes most in detection
6	Latha et al. [23]	COBRE	Skull stripping to remove non brain tissues	Deep Belief Networks (DBNs)	Ventricle region most impactful area for SZs
7	Zhang et al. [22]	COBRE, NMorphCH, BrainGluSchi	Brain affine registration with skull stripping	3D SE-VGG-11BN Model	High Computation cost and time taking

2.2. Summary of existing work

From all the studies done till now we can conclude that 3D CNN does provide overall a better accuracy than any 2D models as it takes the whole brain and extracts the important features from brain MRI images. But the resources, computation power and time required for executing a 3DCNN model is huge.

Also, from previous studies we can determine that 2DCNN can also work quite good in predicting the disease if we can find out region of interest or the most contributing regions, and the increase in dataset can help us to achieve higher and better accuracy with less computation power and within less time.

From paper we can clearly see that subcortical region and ventricular area are the most affecting areas in schizophrenia and the it is most exposed in axial view of the brain which help in predicting disease in more accurate way. And also, different regions of the brain contribute differently in prediction [10], [11], [24]. Also pre-processing of the data plays a vital role in the final prediction as with different preprocessing pipeline accuracy varies a lot. Transfer learning can boost the accuracy of the model when the dataset is very small.

So, we have proposed a model which uses the Axial view of the brain MRI images as the data and Transfer Learning with VGG19 with some Fully connected neural network. It only needs the Axial view of the brain MRI image, and it can predict whether patient is schizophrenic or not.

CHAPTER 3

METHODOLOGY

3.1. System work flow

High level diagram representation of our system is presented in Figure 4. We first did data analysis on each MRI scan to find the correct intensity of axial view of brain image. The data is then preprocessed using threshold and smoothening. The data is then split into training (70%), validation (15%) and testing set (15%). Then this preprocessed data is used to train our Deep Transfer Neural Network model where validation dataset helps us to increase the overall performance.

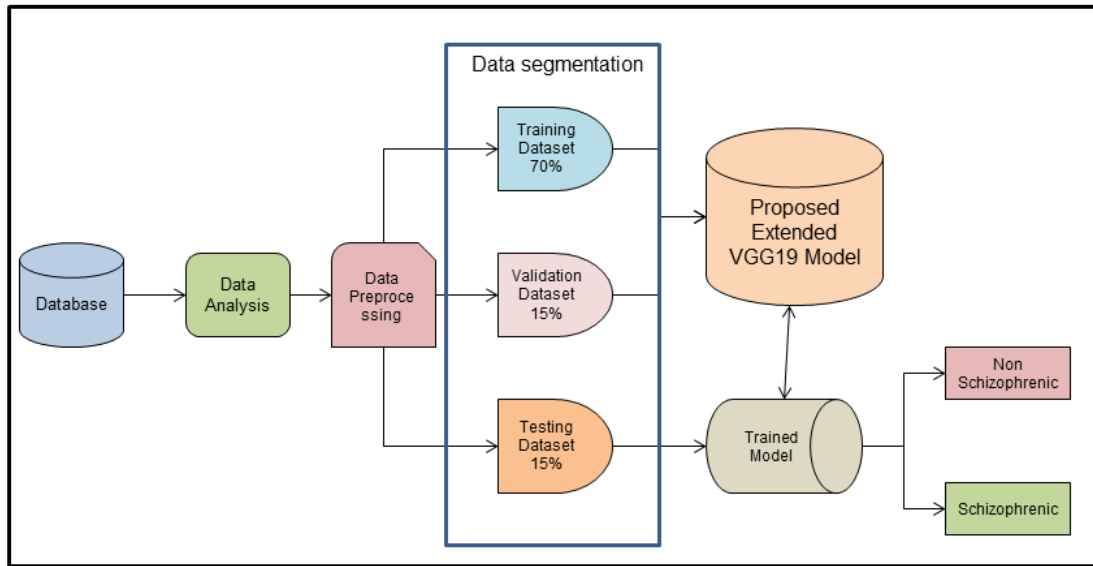


Figure 4 System workflow

We have used 5 different deep learning models on different optimizers with different learning rate and using different activation functions and evaluating performance metrics on all of them. We have also used different preprocessing techniques in order to achieve quality data.

3.2. Dataset

Dataset used in the model is COBRE (The Center for Biomedical Research Excellence) dataset which is a public dataset available on http://fcon_1000.projects.nitrc.org/indi/retro/cobre.html, overview of the dataset can be seen on Table 2. The dataset contains binary classified data of around 148 patients (ages ranging from 18 to 65 in each group), where 74 patients are healthy, and 72 patients are schizophrenic and 2 patients are unknown.

The dataset contains the following data for every participant:

- Resting fMRI
- Anatomical fMRI
- Observable characteristics of data for every participant is provided with following columns:
 - Gender
 - Age
 - Handedness
 - Diagnostic Information

This information is essential to determine further observations on the patients, like which group of age is mostly targeted and which gender is more prone to this disease and the which handedness patient are frequently diagnosed with schizophrenia, where fMRI stands for function Magnetic Resonance Imaging. Every patient is provided with diagnosis values where none determines that the patient is healthy whereas some value determines that the patient is suffering from schizophrenia.

Table 2 Overview of dataset

	Current Age	Gender	Handedness	Subject Type	Diagnosis
40061	18	Male	Right	Control	None
40090	18	Female	Right	Control	None
40046	18	Male	Left	Patient	295.70 depressed type
40002	19	Male	Right	Patient	295.3
40117	19	Male	Right	Patient	295.3
40145	19	Male	Right	Patient	295.6
40113	20	Male	Right	Control	None
40000	20	Female	Right	Patient	295.9
40029	20	Male	Right	Patient	295.6
40133	20	Male	Right	Patient	295.3

Demographics of dataset is given in Table 3, which consists account of male & female, and statistics of dataset with image quality & analysis.

Table 3 Sample characteristics of COBRE dataset

Characteristics	Values
No. of images	146
No. of normal images	72
No. of schizophrenia images	74
Patient demographics	
Age, mean (SD), years	36.97 (12.78)
Age, range (min/max)	18/65
Female, No./total (%)	37/146 (25.34)
Image quality	
Acquisition year	2009 to 2013

Scanner field strength	3T
No, with excessive noise	1
No, with image error	-
Psychiatric diagnosis	
Schizophrenia (broad)	-
Schizophrenia (strict)	74
Schizoaffective disorder	-

3.3. Data analysis

The COBRE dataset consisted of 3D T1 weighted image (the “spin-lattice” relaxation time) with dimension (192*256*256) and weighted scans of the brain. For Successful prediction of schizophrenia, we need compulsorily need brain MRI images, and there are different views such as coronal, axial, sagittal, view, but we particularly need axial view of brain MRI image. We extracted the 2D axial image which is shown in Figure 5, from the brain MRI by certain analysis of frames of images with time and will preprocess this image into a more important format for the model and will train our model according to it.

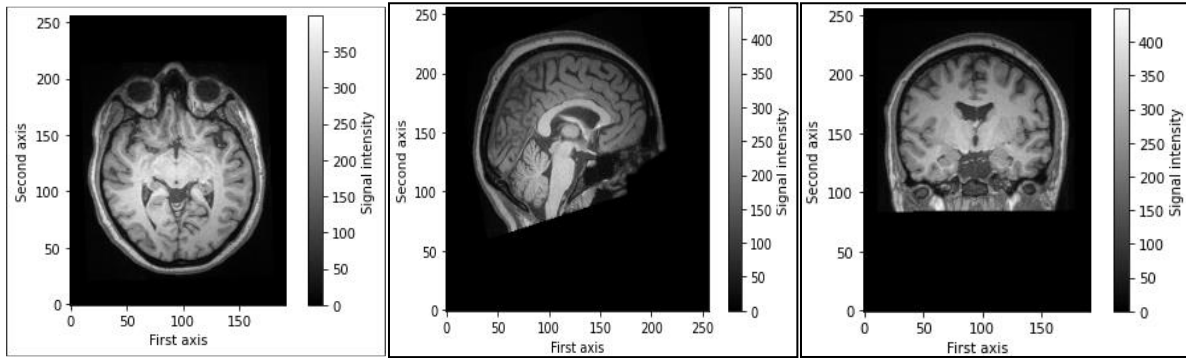


Figure 5 Axial, Sagittal, Coronal views of brain MRI respectively

So, before using the data we did some preprocessing on the 3D images to get the 2D axial view of the brain [21]. Figure 6 shows the subcortical region and ventricular area which contributes the most in schizophrenia detection [22], and in axial view it is most exposed. NIfTI (Neuroimaging Informatics Technology Initiative) which is a file format for MRI images, they have different slices of brain at different point of time, so for the axial view we need to manipulate the Z axis of the image with time to get correct view of the brain which can help us to predict the disease. So, we tried 36 different time frames for the Z axis of the image which has dimension 256, so the images are formed at an interval of 7 frames and analyzed the data. After analyzing the data, we chose the best interval at which the axial view of the brain was accurate and clearly visible and intensified.

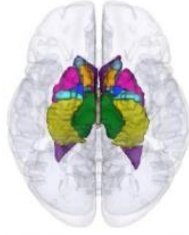


Figure 6 Subcortical region of brain in axial view [22]

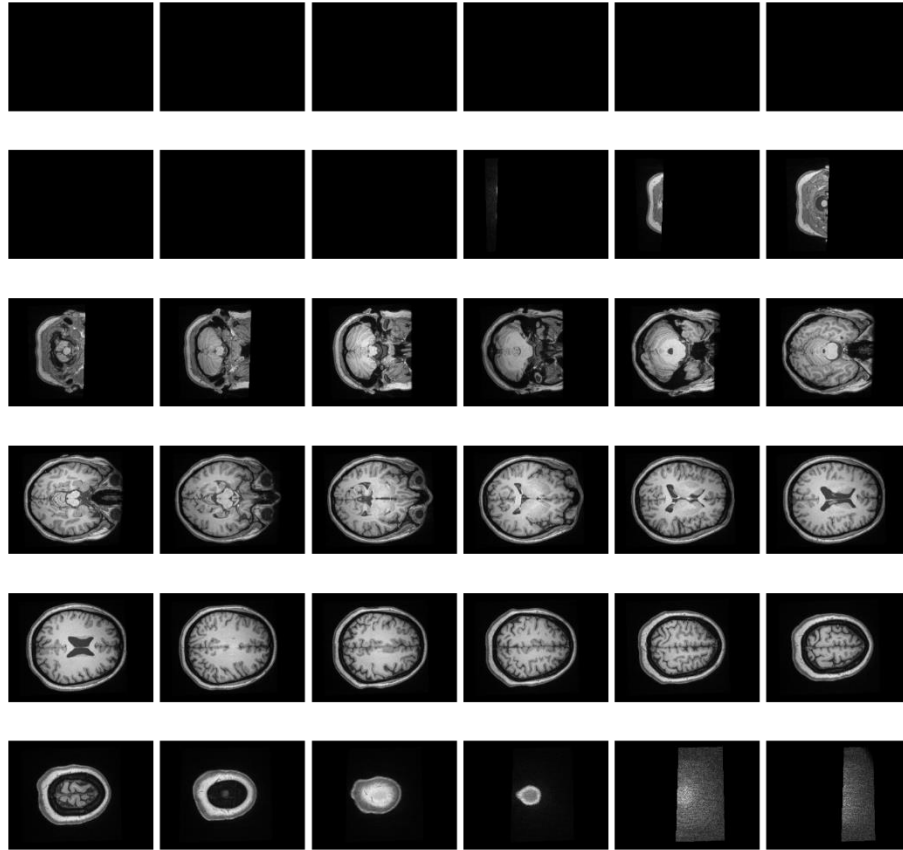


Figure 7 Images of brain at different time intervals

3.4. Data preprocessing

In the preprocessing part we applied binary threshold in the image, so that model gets more distinction between different brain features, also tried different types of smoothing like gaussian blur, median blur, bilateral Filter, blur, with a 5x5 matrix, to add noise to the data in order to make the model more robust, preprocessing pipeline is shown in the Figure 8.

Now since our dataset was very small model doesn't get trained well to predict on new data, so to increase our dataset, instead of 1 image per patient we took 7 images per patient for training and validation purpose whereas testing was done only on 1 image per patient. We also tried data augmentation by applying different filters on the same data to generate more data. So, this made our training and validation data 7 times larger which helped to increase the accuracy and robustness of the model.

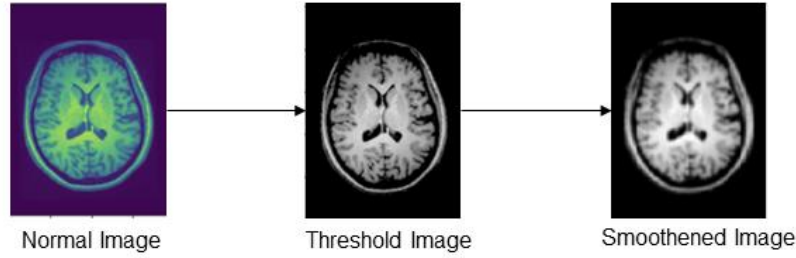


Figure 8 Preprocessing pipeline

3.5. Data segmentation

After preprocessing the data, we divide the data into training, validation and testing set with 70%, 15%, 15% distribution using stratify technique to balance both positive and negative labels in all the sets. Training data contains 102 patient's data with each patient's 7 slices of brain images are used so total training data to train our model is 714 images and validation data contains 21 patient's data with 7 slices each so total validation set is 147 brain MRI images and testing data contains 21 brain MRI images of patients with 1 slice each so total testing data is 21.

3.6. Proposed extended VGG-19 model

We have proposed extended VGG19 model which is a sequential model where VGG19 model is used for transfer learning and further extended by 4 deep neural layers for better learning and prediction.

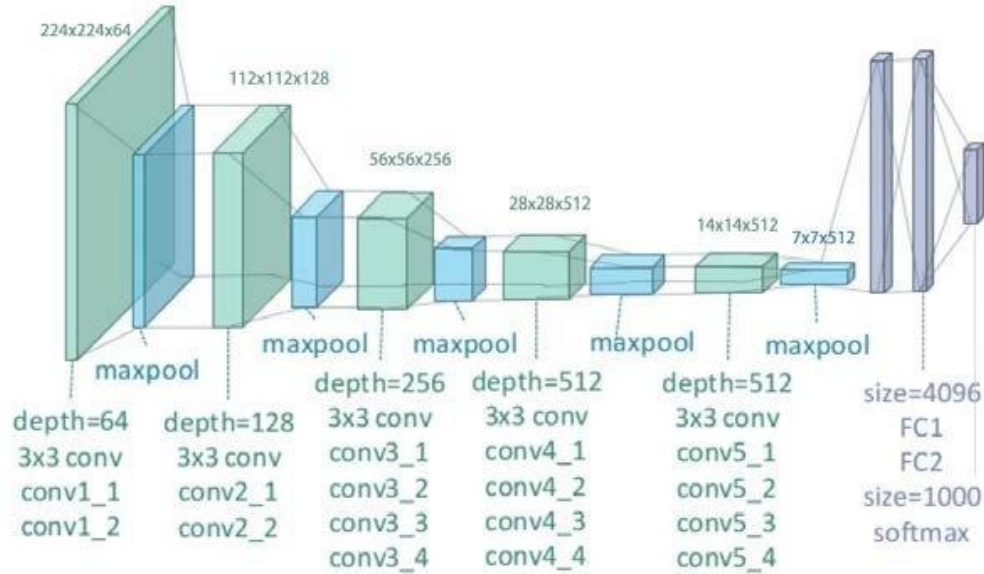


Figure 9 VGG-19 architecture [25]

VGG-19: VGG19 is a variant of VGG model which in short consists of 19 layers (16 convolution layers, 3 Fully connected layer, 5 Max Pool layers and 1 SoftMax layer) the architecture of VGG19 is shown in Figure 9. Each layer has a number of feature maps, each feature map has a number of neurons, and the convolution filter extracts the input features. The parameter sharing mechanism of the convolutional layer greatly reduces the number of parameters. Max Pooling is used to down sample the dimension of the input features. VGG19 is being used as transfer learning with

pretrained ImageNet weights model to optimize and improve the network. The input layer size of VGG16 is 192×256 , and a stack of convolutional layers and pooling layers make up the convolution module.

The convolutional layer is expressed as follows:

$$y_j^s = f \left(\sum_{i \in M_j} y_i^{s-1} * t_{ij}^s + a_j^s \right). \quad (1)$$

In Equation (1), let's assume that input layer or pooling layer is represented by $s - 1$ where s is the convolutional layer, then y_i^s is the s convolutional layer's j^{th} feature map; the right hand side of the Equation (1) shows the $s - 1$ layer's feature map. Convolutional operation is performed with the j^{th} convolution kernel t_{ij}^s of the s layer and sum where a represents the bias and $f(\cdot)$ represents the ReLU activation function.

The pooling layer, which fulfils the job of scaling dimensionality, immediately follows the convolutional layer. The following is the calculation equation:

$$y_i^s = f(\alpha_j^s \text{down}(y_i^{s-1}) + a_j^s). \quad (2)$$

In Equation (2), The pooling function, $\text{down}(\cdot)$, finds the highest value possible for a particular area of the feature map; α_j^s and a_j^s , reflect the bias and weight of the pooling, respectively.

On one hand, the image size is decreased and the amount of model calculation is regulated by working together with the convolutional layer and the pooling layer. On the other hand, the convolution data of the large receptive field is obtained indirectly, and the high-dimensional feature map is obtained.

Extended VGG-19: After getting the feature processed in all the layers of VGG19 we skip the last layer and extend the model with our modification, one layer to flatten the output given by VGG19 which was pretrained on ImageNet dataset whose weights are not allowed to train, and followed by a dropout layer with 30% of closed neurons and 3 Fully connected hidden layers with number of neurons 128, 64, 32 respectively using ReLU (Rectified Linear Unit) as the activation function and after first two layers to avoid overfitting we have added a dropout layer with 30% of closed neurons so that our model learns more important and contributing features from the images brain MRI images. After hidden layers we have added an output layer which predicts the final output of the model it uses sigmoid as activation function. Complete modified VGG19 architecture with extra layers can be seen in Figure 10. We utilize the sequential model of the keras library because we want to fine-tune our hyperparameters so that we may give each predictor a unique set of weights. Modified VGG19 which contains 20,024,384 parameters which are non-trainable and further 1 flatten layer followed by 3 fully connected neural network layers which contains 3,156,225 trainable parameters. We added 3 dropout layers in the model after each fully connected layer. Sequential model details with layer size and parameter information are shown in Figure 11. We have used Adam function as our optimizer for minimizing the loss using Learning rate α , using Binary Cross entropy as our loss function, which was trained on 100 epochs with using Early

Stopping callback functionality which prevents overfitting and stops the training of the model once loss has converged.

$$\alpha = 10^{-3} \quad (3)$$

$$Loss = -(y \log(p) + (1 - y) \log(1 - p)) \quad (4)$$

Equation (3) shows the value of the learning rate which was used in optimization of loss while training the model.

Equation (4) shows the binary cross entropy loss function equation which is being minimized by training the model where y shows actual label of the data and p shows the predicted probability by our model.

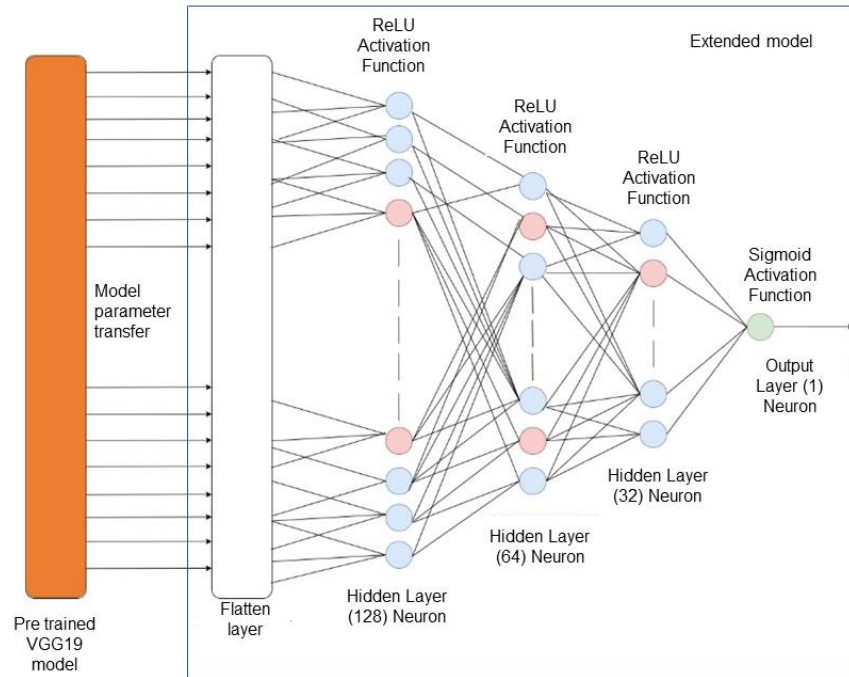


Figure 10 Proposed extended VGG19 model for schizophrenia detection

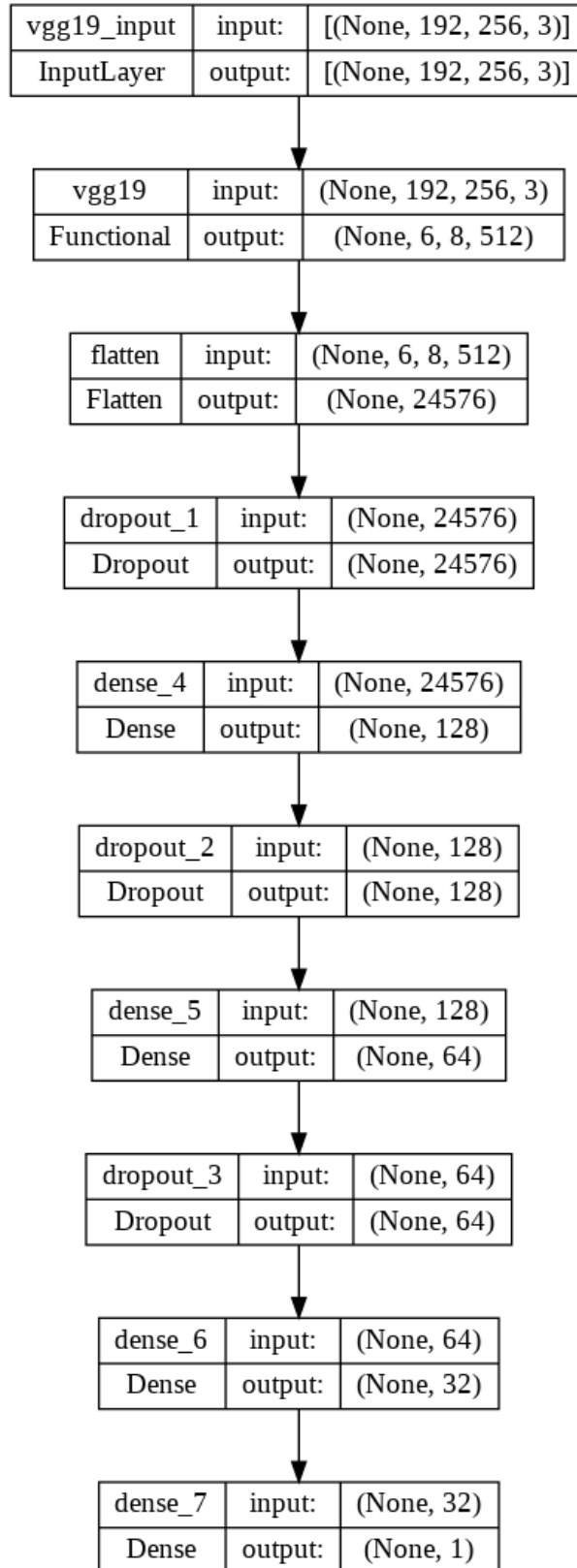


Figure 11 Model summary of proposed extended VGG19 model

Transfer learning: Transfer learning addresses the drawbacks of deep learning, which need a lot of sample training models. It is possible to train a model with fewer data sets by using a pretrained model that has been trained on a sizable amount of data. A training technique called fine-tune keeps the model's feature extraction layer while retraining the model's classification layer. The feature extraction layer of the pretraining model is fixed, and it uses the VGG19 network that has been pretrained on the ImageNet data set where weights of the model are not allowed to train and the knowledge of the model is transferred to our extended layers which utilizes the knowledge as well as image inputs. Transfer learning model which is being used is shown in Figure 12. To finish training the schizophrenia classification model, we train it on VGG19 and followed by 4 layered DNN to predict the final output.

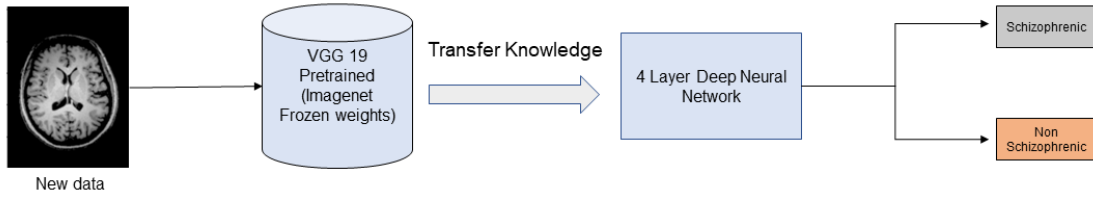


Figure 12 Transfer learning training process

Evaluation metrics: We employ evaluation metrics including precision, recall, accuracy, and F1 Score that are frequently utilized in classification tasks. The confusion matrix in Table 2 serves as an illustration of the categorization task. True positive (TP) denotes that the model actually predicted the number of sample positive classes and that the positive class is projected to be a positive class. False negative (FN) signifies that the positive class is projected to be a negative class and that the model actually predicted the number of negative classes in the sample. False positive (FP) means that the negative class is projected to be a positive class and that the model actually predicted the number of positive classes of samples. True negative (TN) denotes that the model accurately predicted the number of sample negative classes and that the negative class is predicted to be a negative class.

Table 4 uses the confusion matrix to demonstrate the classification task.

Table 4 Confusion matrix

	Predicted positive class	Predicted negative class
Actual positive class	TP	FN
Actual negative class	FP	TN

$$Recall = \frac{TP}{TP + FN} \quad (5)$$

Equation (5) shows the calculation of recall which indicates how many actual positive was correctly identified.

$$Precision = \frac{TP}{TP + FP} \quad (6)$$

Equation (6) shows the calculation of precision which indicates how many positive outcomes were predicted correctly.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (7)$$

Equation (7) shows the calculation of accuracy which indicates the fraction of outcomes model predicted correctly.

$$F1\ Score = \frac{TP}{TP + \frac{1}{2}(FP + FN)} \quad (8)$$

Equation (8) shows F1 score which combines the recall and precision metric into one for easier evaluation.

CHAPTER 4

EXPERIMENTAL RESULTS

4.1. Experiment system configuration

The proposed extended VGG-19 model was trained and tested on google colab [26]. Virtual machine provided uses Intel(R) Xeon(R) CPU @ 2.30GHz, it has 2 cores with 2 threads per core. RAM available was 12GB with 25GB disk space. Colab provides Nvidia T4 GPU with 16GB memory, and 8.1 TFLOPs 10-12 floating-point operations per second. Packages used for the training of the model are TensorFlow, Keras, Matplotlib, ScikitLearn, nibabel, OpenCV.

4.2. Dataset for evaluation

COBRE dataset is being used which is a 3D MRI image data., Dataset is divided into training, validation and testing set where training and validation was used to train the model whereas testing set will be used here for the evaluation of model. Testing dataset contains 21 set of brain MRI images which 10 images are schizophrenic patient, and 11 images are of non-schizophrenic patients. Test data is analyzed and preprocessed in similar way but only 1 slice of axial view of brain MRI image is taken for evaluation.

4.3. Experimental setup

In this experiment we will try different state of the art classification deep learning model for transfer learning. Further we will experiment on proposed extended VGG19 model using different hyper parameters like activation function and preprocessing technique and comparative study between them. Next, we will discuss the results with existing state of the art model using same dataset. After this we will briefly study the comparison between all the model implemented with the help of performance metrics such as accuracy, precision, recall, F1 score.

4.4. Performance of deep learning models

We have studied various deep learning models which have performed good on image classification in past like VGG, ResNet, Xception, EfficientNet, there were more variation of these model. We have experimented on VGG16, VGG19, ResNet101 and EfficientNetB0 in order to get better results in predicting schizophrenia.

4.4.1. Transfer learning using VGG16

In the previous State of the art model transfer learning was used with VGG16 model, so we trained our dataset using VGG16 and the Figure 13 (a) shows the ROC curve of VGG16 model with x axis representing false positive rate and y axis representing true positive rate. We were able to achieve an accuracy of 0.71 with test loss 0.86. The F1 score, Precision, recall value were 0.63, 0.66, 0.63 respectively. Further Figure 13 (b) shows the graph of training loss and validation loss with number of epochs which shows that with number of epochs the validation and training loss is decreasing at some point validation and training loss spikes it is due to outliers. ReLU activation function was used with Adam optimizer, using early stopping callback to reduce overfitting.

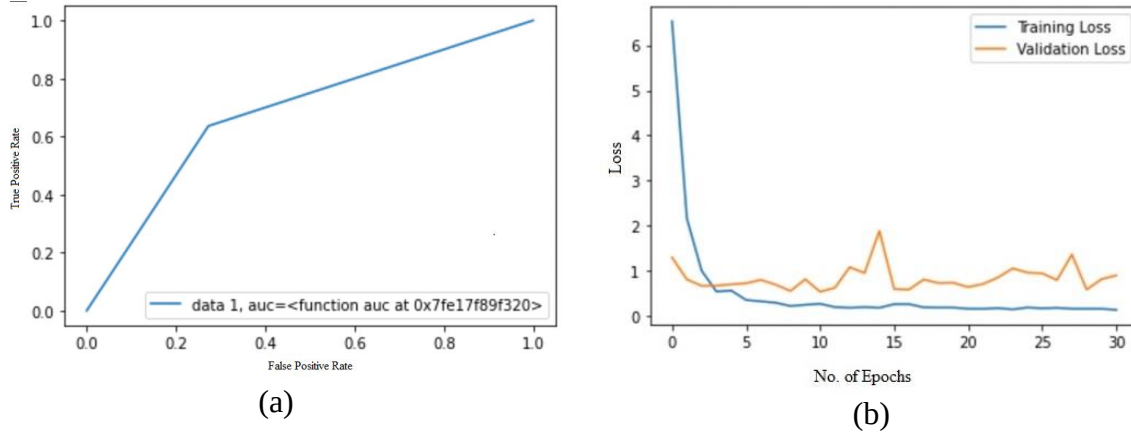


Figure 13 (a) ROC curve for VGG16 (b) Loss curve for VGG16

4.4.2. Transfer learning using ResNet101

ResNet50 has performed very well on image classification, and ResNet101 is upgraded version of ResNet50 which has performed better than ResNet50 in almost all scenarios. So, we trained our dataset using ResNet101. Figure 14 (a) shows the ROC curve on using ResNet101 model with x axis representing false positive rate and y axis representing true positive rate. We were able to achieve an accuracy of 0.77 with test loss 0.36. The F1 score, Precision, recall value were 0.76, 0.72, 0.72 respectively. Further Figure 14 (b) shows the graph of training loss and validation loss with number of epochs which shows that with number of epochs the validation and training loss is decreasing at some point validation and training loss spikes it is due to outliers. Hyperparameters used in training were fine-tuned and kept same as it was in last two models.

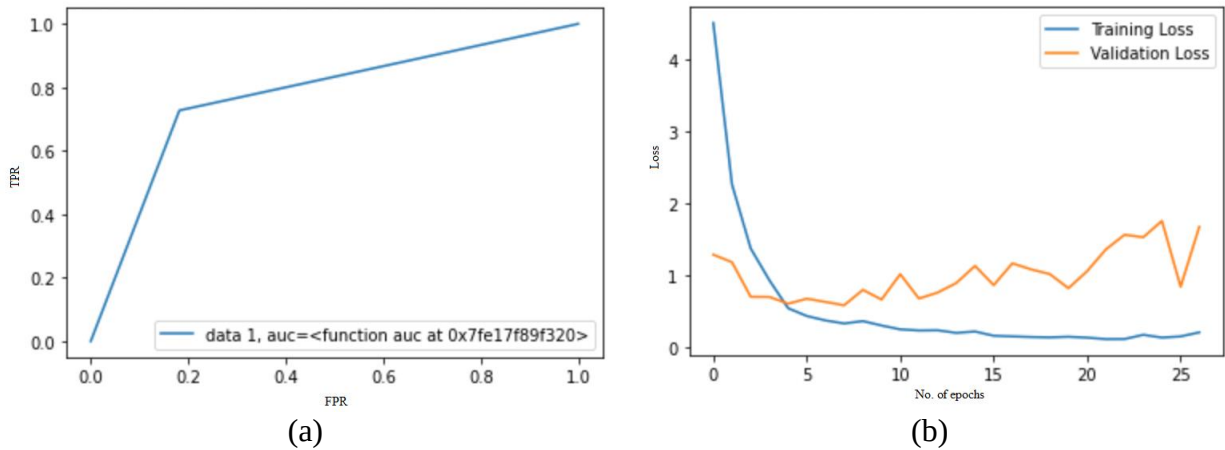


Figure 14 (a) ROC curve for ResNet101 (b) Loss curve for ResNet101

4.4.3. Transfer learning using EfficientNetB0

EfficientNet performs well in terms of transfer and achieves cutting-edge accuracy on the CIFAR-100. So, we trained our dataset using EfficientNetB0 and it helped us to achieve maximum accuracy of 0.81 which was more than VGG16 and ResNet101 with test loss 0.47.

Figure 15 (a) shows the ROC curve of the EfficientNetB0 with x axis representing false positive rate and y axis representing true positive rate. The F1 score, Precision, recall value were 0.81, 0.81, 0.78 respectively. Further Figure 15 (b) shows the graph of training loss and validation loss with

number of epochs which shows that with number of epochs the validation and training loss is decreasing at some point validation and training loss spikes it is due to outliers. Same hyperparameters were used as described above.

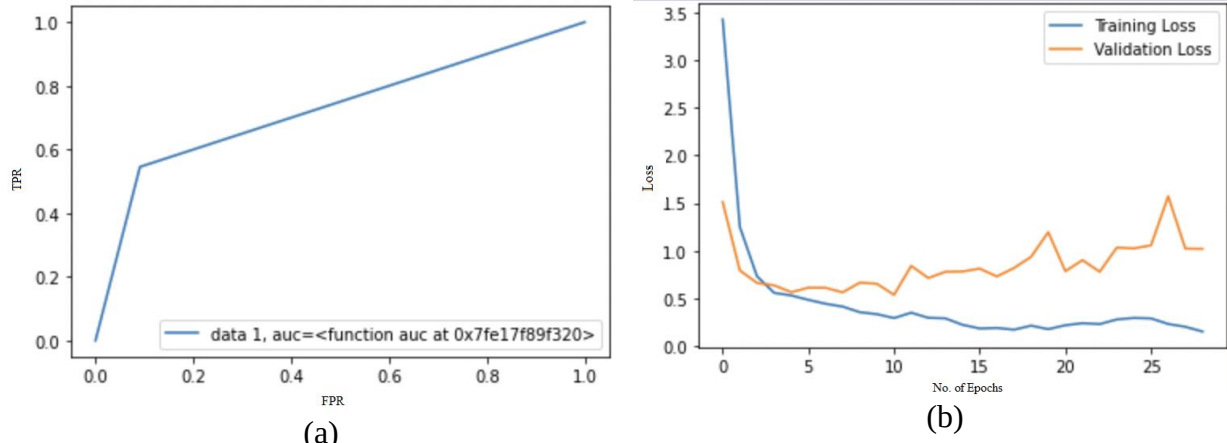


Figure 15 (a) ROC curve for EfficientNetB0 (b) Loss curve for EfficientNetB0

4.4.4. Proposed extended VGG19 model

VGG19 is the new evolved version of VGG16 in which 19 layers are used to train the model while in VGG16 there are 16 layers. So, we trained our dataset using VGG19 model and it helped us to achieve accuracy more than the model which was given in the paper. Figure 16 (a) shows the ROC curve of the VGG19 with x axis representing false positive rate and y axis representing true positive rate. Maximum accuracy achieved is 0.91 with test loss 0.23. The F1 score, precision, recall value were 0.90, 0.90, 0.916 respectively. Figure 16 (b) shows the graph of training loss and validation loss with number of epochs which shows that with number of epochs the validation and training loss is decreasing at some point validation and training loss spikes it is due to outliers.

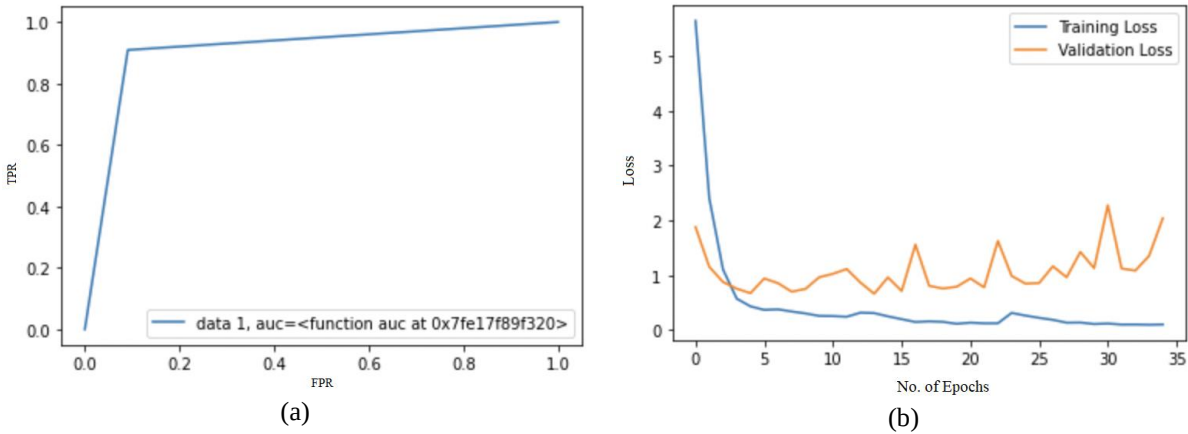


Figure 16 (a) ROC curve for extended VGG19 model (b) Loss curve for VGG19 model

4.4.5. Comparison of different hyperparameters

We tried different parameters for achieving best results, for activation function we used Tanh function which gave 0.77 accuracy with precision 0.81, F1 score 0.78, and Recall 0.81, then we tried using sigmoid activation function which gave us 0.71 accuracy with precision 0.66, F1 score

0.63, and Recall 0.63. The best result was achieved in **ReLU** activation function which gave us 0.91 accuracy with precision 0.90, F1 score 0.90, and Recall 0.92.

Other than activation function we tried filtering methods in order to obtain best results, Gaussian blur was used which gave accuracy of 0.72 with precision 0.81, F1 score 0.75, and recall 0.81. **Average blur** was used which gave us best results with precision 0.90, F1 score 0.90 and recall 0.92.

Table 5 shows the comparison of different hyperparameters used in training to get the best results. First row shows different activation function used in hidden layers of the model and their results respectively. Second row shows the different way of filtering which is used to reduce overfitting.

Table 5 Comparative performance of hyperparameters

	Hyperparameter	Accuracy	Precision	Recall	F1-score
Activation Function	ReLU	0.91	0.90	0.91	0.90
	Tanh	0.77	0.81	0.81	0.78
	Sigmoid	0.71	0.66	0.63	0.63
Filter for image processing	Gaussian Blur	0.72	0.81	0.81	0.75
	Average Blur	0.91	0.90	0.91	0.90

4.5. Comparison with existing models

We will compare our model with the existing model given in the paper which used VGG16 as transfer learning and modified the VGG16 model by decreasing layers, whereas in our model we have used VGG19 followed by 4 Layer model using ReLU activation function in hidden layers and sigmoid at the output layer.

Table 6 shows the difference in performance metrics of proposed model and model given by Zheng et al. [21] and Latha et al. [23], clearly by seeing the performance metrics we can say that our model has outperform the existing model.

Table 6 Comparison of performance metrics

	Accuracy	Precision	Recall	F1-score
Zheng et al. [21]	0.87	0.87	0.890	0.85
Latha et al. [23]	0.90	0.93	0.875	0.89
Proposed extended VGG19 model	0.91	0.90	0.916	0.90

4.6. Comparison of performance of all models

Now, we will compare all the Deep Neural Network (DNN) models discussed above based on accuracy, precision, recall, and F1 score. Figure 17 shows the comparison of accuracies of all the experimented models, the proposed model has 18%, 13%, and 9% higher accuracy than VGG16, ResNet101, and EfficientNetB0 experimented models respectively and the proposed model has

3% and 0.91% higher accuracy than the Zheng et al. [21] and Latha et al. [23] existing models respectively. Here we can see that the proposed model is outperforming every other model including the existing one.

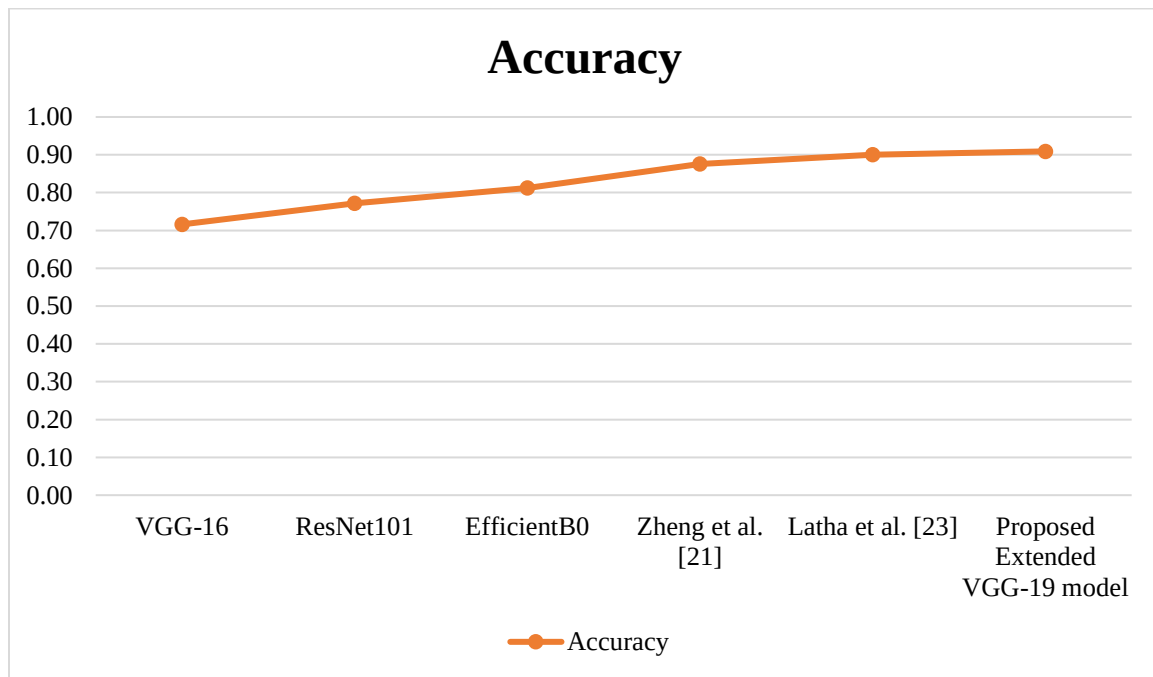


Figure 17 Comparison of accuracy

Figure 18 shows the comparison of the precision of all the experimented models, the proposed model has 24%, 18%, and 9% higher precision than VGG16, ResNet101, and EfficientNetB0 experimented models respectively and the proposed model has 3% higher precision than the Zheng et al.[21] and 3% lower than Latha et al.[23] existing models. Here we can see that the proposed model is outperforming every other model including Zheng et al. [21] except Latha et al. [23]

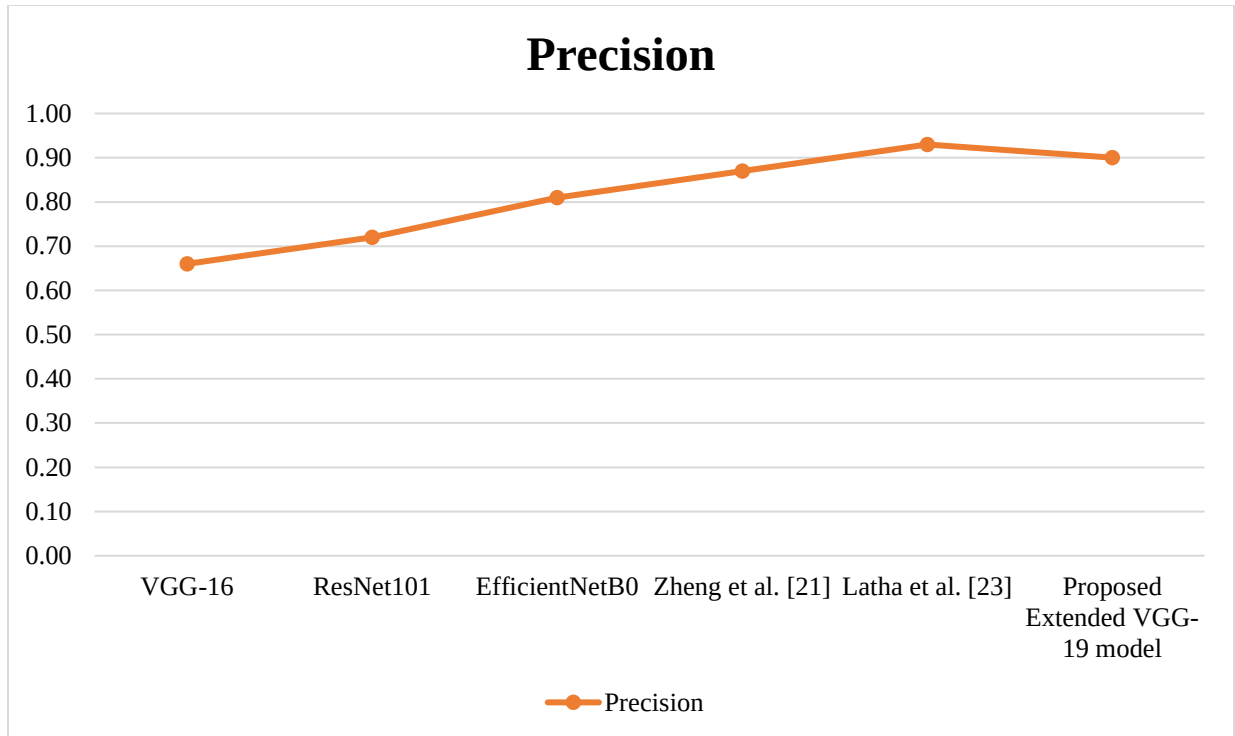


Figure 18 Comparison of precision

Figure 19 shows the comparison of recall of the entire experimented model, the proposed model has approximately 29%, 20%, and 14% higher recall than VGG16, ResNet101, and EfficientNetB0 experimented models respectively and the proposed model has approx. 3% and 4% higher recall than the Zheng et al. [21] and Latha et al. [23] existing models respectively. Here we can see that the proposed model is outperforming every other model including the existing one.

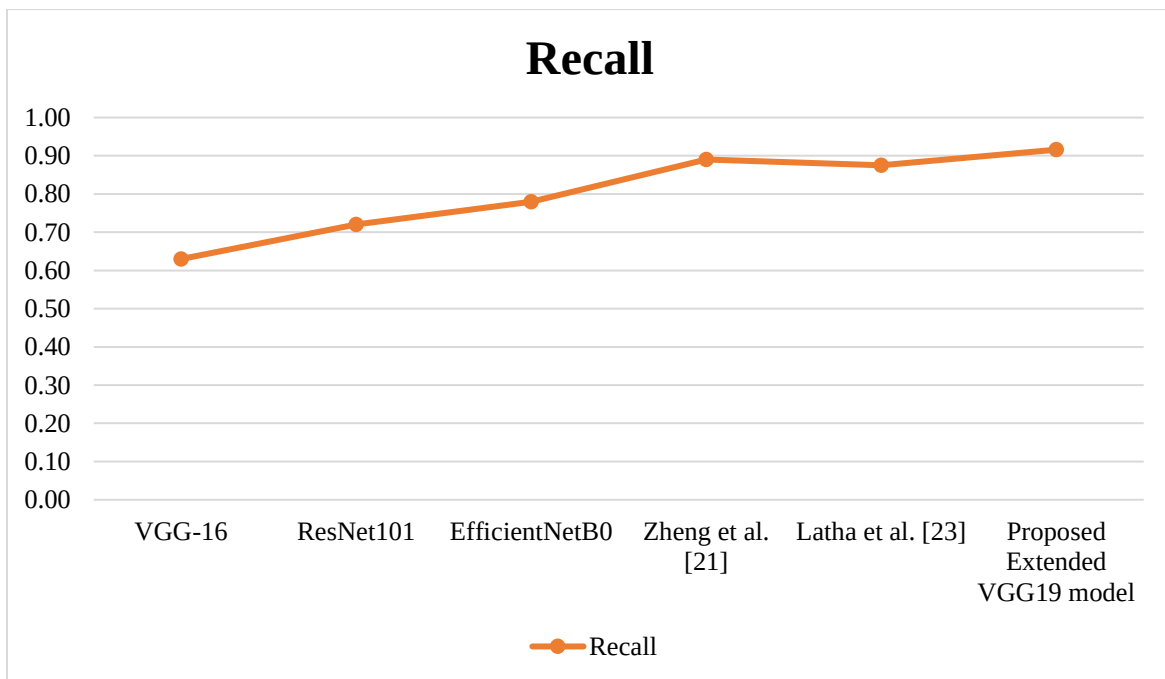


Figure 19 Comparison of recall

Figure 20 shows the comparison of the F1 score of the entire experimented model, the proposed model has approx. 27%, 14%, and 9% higher F1 score than VGG16, ResNet101, and EfficientNetB0 experimented models respectively and the proposed model has approx. 5% and 1% higher F1 score than the Zheng et al.[21] and Latha et al. [23] existing models respectively. Here we can see that the proposed model is outperforming every other model including the existing one.

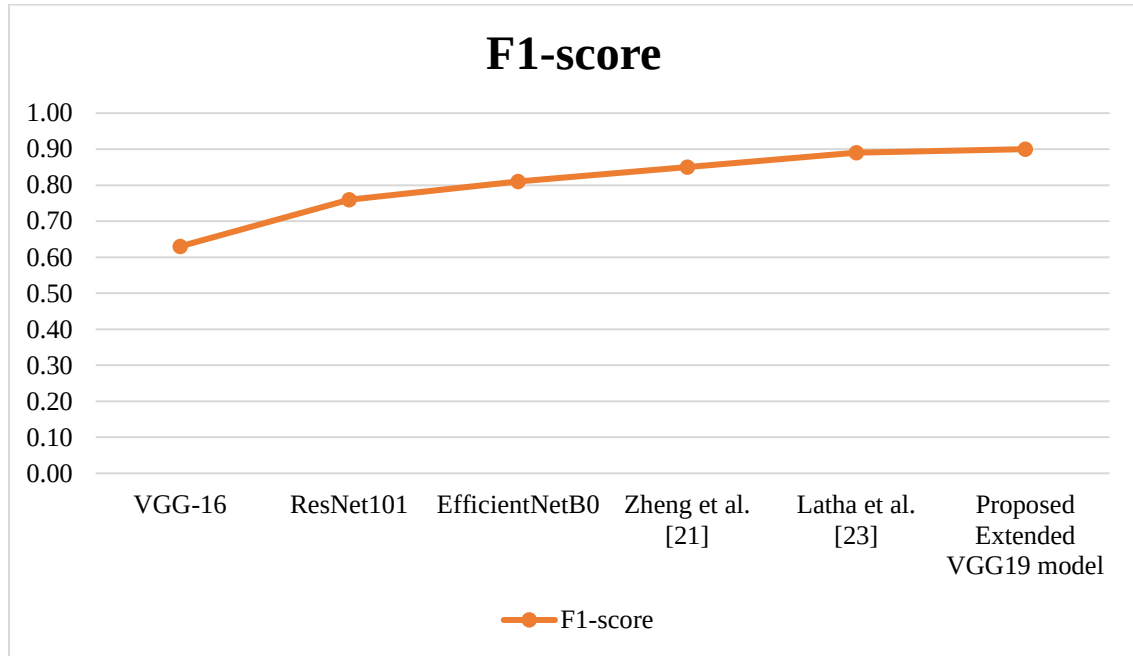


Figure 20 Comparison of F1-score

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

5.1. Conclusion

Previous models using deep learning model which needs a large amount of data, also all the slices of the brain image MRI were used which consist of all views which were creating noise in the data as some unnecessary information was going in the model which reduced the performance of the model. So we proposed a model which took the unprocessed 3D scans of the brain and analyzed the data to get only the axial view of image as subcortical region and ventricular areas are most exposed in it and they are most contributing factor in the case of schizophrenia which solves the unnecessary data or noise data problem. After analyzing the data we passed them through the preprocessing pipeline to threshold the image and apply image augmentation. We implemented the model based on transfer learning with the VGG-19 pretrained model and used the training parameters for our dataset to handle the large data problem. We then transfer the knowledge to fully connected layers which learn and predict whether the person has schizophrenia or not. This project also compares the results of different convolution models available which is pre trained on ImageNet dataset. The current best existing model in 2D CNN used VGG-16 for getting parameters and then training the data on an improved classifier. We compared this model with the current best 2D CNN model and observed an increase in accuracy by maximum 3% and minimum 1%. When compared to 3D CNN based models, we achieved a faster performance using our model with minimal loss in accuracy. We compared the performance of our proposed model over COBRE dataset. For the comparison of different models, we used evaluation metrics like accuracy, precision, recall, and F1-Score. Evaluation results indicate that our proposed model performs faster and better than the existing models for schizophrenia detection using MRI.

5.2. Future scope

As the publicly available data of MRI scans is very low, the accuracy and evaluation metrics of all implemented models can be improved by increasing the data. Also, for improvement in accuracy, there can be another model combined with VGG-19. Another scope of improvement is by using different phenotype data with MRI scans.

Also, this classification model can be used for other brain disorders which take brain MRI images for detection.

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